Chapter 5
Determination of Forward and Futures Prices

Chapter 2 established that the futures price converges to the spot price at delivery. This chapter answers the question that leaves open: what determines it *before* delivery?

The answer, for a large class of assets, is remarkably tight. The forward price is not a forecast, not a consensus view, and not the result of supply and demand for the contract. It is pinned by arbitrage to a specific relationship with the spot price, and a trader who quotes anything else can be taken apart by someone with a bank account and no opinion whatsoever about the future direction of the underlying price.

That is the good news. The rest of the chapter is about where the argument weakens: assets that cannot be sold short, assets held for use rather than investment, and the fact that the arbitrage requires a balance sheet, which is not free, a theme that recurs through Chapters 6, 7, and 8.

What arbitrage means. An **arbitrage opportunity** is a trading strategy that costs nothing to set up, can never lose money, and makes money in at least one state of the world; equivalently, it pays money today and can never lose thereafter. The argument of this chapter, and the pricing in every chapter after it, rests on the **no-arbitrage assumption**: such opportunities cannot persist, because anyone who found one would trade it in unlimited size and prices would move until it was gone. It is an assumption about prices, not about people. It does not say that arbitrages never appear, only that the market does not leave them lying about, and Section 5.15 examines where it fails in practice.

Section 5.1 draws the distinction the chapter turns on, between assets held for investment and assets held for use, and sets out what short selling requires. Sections 5.2 to 5.5 are the arbitrage argument, its extensions to known income and known yield, and the value of a forward already on the books. Sections 5.6 to 5.8 apply it to the three markets where it is tested daily (index futures, currencies, and commodities), and Sections 5.9 to 5.11 collect the pieces into the cost of carry, the implied repo rate, and the roll yield of a position held across contracts. Sections 5.12 and 5.13 settle two questions that are often confused: whether forward and futures prices differ, and whether either forecasts the spot price. Section 5.14 shows how the shape of the curve is traded, and Section 5.15 is where the arbitrage argument stops.

5.1 INVESTMENT ASSETS, CONSUMPTION ASSETS, AND SHORT SELLING

The arbitrage arguments that follow depend on a distinction.

An **investment asset** is held by a significant number of investors purely for investment (gold, silver, stocks, bonds, currencies). It need not be held *only* for investment; gold is also used industrially. What matters is that enough holders are indifferent between holding the physical asset and holding a claim on it, so that arbitrage can proceed by borrowing the asset from them.

A **consumption asset** is held primarily to be used: oil, copper, wheat, live cattle. No one holds crude oil as a store of value. Its holders are precisely the negation of the condition above: they are *not* indifferent between the physical asset and a claim on it, because what they want from it is the use. A refinery needs barrels in the tank to keep the plant running, and a claim on barrels next month will not run it.

Two consequences follow, and the second matters more than the first. Such a holder will not lend the asset, so the short sale that the arbitrage of Section 5.2 depends on is simply unavailable: you cannot readily borrow oil from a refinery that intends to process it next week. But even a holder who owns the asset outright, and needs to borrow nothing from anyone, will not *sell it now and buy it back later*, however high today's price, because the whole point of holding it is to have it in the meantime. The trade that would keep a forward price from falling too far below what it costs to hold the asset until delivery is therefore not merely hard to finance: no one who has the asset wants to do it. That is what turns this chapter's equalities into one-sided inequalities for commodities, and it is the economic content of the convenience yield in Section 5.8.

Short selling. To sell short, an investor borrows the asset from a holder, sells it, and later buys it back to return. The mechanics matter for pricing. The short seller must pass through any income the asset generates (dividends, coupons) to the lender, and pays a **borrowing fee** for the loan. For liquid stocks that fee is a few basis points; for hard-to-borrow stocks it can be tens of percent, and the position may be recalled at any time.

Every result in this chapter assumes short selling is costless where it is possible. Where it is not possible at all, the arbitrage works in only one direction, and the results become inequalities rather than equalities, as Section 5.8 shows.

Notation and assumptions. Throughout, S_0 is today's spot price, S_T the spot price at delivery, F_0 the forward or futures price, T the time to delivery, and r the continuously compounded risk-free rate for maturity T (Section 4.7). The standing assumptions are that participants face no transaction costs, are subject to the same tax rate, can borrow and lend at r, and take arbitrage opportunities when they arise.

The third of these deserves skepticism from the outset: no one borrows at the risk-free rate. We keep the assumption because it makes the structure clear, and return to what it costs in Section 5.15.

5.2 THE BASIC ARBITRAGE

Consider an investment asset providing no income and costing nothing to store (a non-dividend-paying stock is the standard example). Then

$$\boxed{F_0 = S_0 e^{rT}} \tag{5.1}$$

The argument is worth setting out in both directions, because the pair of trades is the template for everything that follows. Each direction is an arbitrage opportunity in the sense defined at the start of the chapter: it costs nothing, cannot lose, and makes money if the forward price is out of line.

If $F_0 > S_0 e^{rT}$, borrow S_0 at rate r, buy the asset, and short a forward contract. At T, deliver the asset for F_0 and repay $S_0 e^{rT}$. The profit is $F_0 - S_0 e^{rT} > 0$, with no investment and no risk. This is the **cash-and-carry** trade: borrow cash, buy the asset, carry it to delivery. It is the trade the rest of the book refers to by that name.

If $F_0 < S_0 e^{rT}$, short the asset for S_0, invest the proceeds at r, and take a long forward. At T, receive $S_0 e^{rT}$ from the investment, pay F_0 for the asset under the forward, and return it to

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the lender. The profit is $S_0 e^{rT} - F_0 > 0$. This is the **reverse cash-and-carry**, and Section 5.1's question of who will part with the asset is the question of who can do this one.

Since both cannot be available, equality must hold.

Example 5.1 A forward on a non-dividend-paying stock

A stock trades at \$50, the six-month risk-free rate is 4%, and the stock pays no dividend. From (5.1) the six-month forward price is

$$F_0 = 50\, e^{0.04\times 0.5} = 50 \times 1.020201 = \$51.01$$

Suppose a dealer quotes \$52. Borrow \$50, buy the stock, short the forward. Six months later, deliver the stock for \$52 and repay \$51.01, keeping \$0.99 per share, with no capital committed and no exposure to where the stock went.

Suppose instead the quote is \$50. Short the stock, invest the \$50 to receive \$51.01, and buy the stock back at \$50 under the forward. Profit \$1.01.

No view on the stock is required, and no probability distribution enters. The forward price is determined by the spot price and the interest rate alone, which is why Chapter 2's remark that a futures price is not a forecast is more than a caution.

The constant rate is a convenience, not an assumption of the argument. Financing the purchase by selling zero-coupon bonds maturing at T, rather than by borrowing at r, the same two trades show that

$$F_0 = \frac{S_0}{P(0,T)} \tag{5.2}$$

where $P(0,T)$ is today's price of a zero-coupon bond paying \$1 at T. This holds whatever rates do in between; only the bond's traded price enters, not a model for it. A constant rate makes $P(0,T) = e^{-rT}$, and (5.2) is (5.1) again. Section 28.7 starts from (5.2), in an economy where rates are stochastic by assumption.

When short selling is costly

Example 5.1 used both arbitrages, and the second borrows the stock. It is natural to expect that an expensive borrow degrades the equality into a band, and eventually it does, but not for the reason it first appears, and not as readily.

The lower bound does not need a short sale at all. If $F_0 < S_0 e^{rT}$, an investor who *already owns* the asset can sell it, invest the proceeds at r, and take a long forward. At T that investor has the asset back, having received $S_0 e^{rT}$ and paid F_0, and is better off than by holding throughout. Nothing has been borrowed and no fee has been paid. This is exactly why an investment asset was defined in Section 5.1 by holders who are indifferent between the asset and a claim on it: their willingness to exchange one for the other is what defends the lower bound, and the cost of borrowing is nothing to them.

A borrowing fee therefore does not by itself widen the no-arbitrage range. What widens it is holders who will not or cannot make that exchange: an index fund mandated to hold the constituents, an insider under lockup, a taxable holder who would realize a gain by selling, or a holder already lending the shares and collecting the fee, for whom selling means surrendering that income. Where the sale route is closed and only borrowed stock is available, write b for the continuously compounded borrowing fee. The cash-and-carry trade is unaffected, so the upper bound stands; the reverse trade surrenders b to the lender, and

$$S_0\, e^{(r-b)T} \;\le\; F_0 \;\le\; S_0\, e^{rT} \tag{5.3}$$

It is worth being clear about which case is which. If the fee accrues to the *marginal* holder (the one whose readiness to sell would close the gap), the asset is an investment asset paying

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income at rate b, and Section 5.4 gives the equality $F_0 = S_0 e^{(r-b)T}$, not a band. Equation (5.3) is the intermediate case: some holders earn the fee, but the holder who would otherwise arbitrage cannot act. Example 5.2 puts numbers on the band.

Example 5.2 A hard-to-borrow stock

A stock trades at \$50 with a six-month rate of 4%. For an easy-to-borrow stock, Example 5.1 pins the forward at \$51.01.

Now suppose the stock is heavily shorted and the borrowing fee is 3% a year. From (5.3) the bounds are

$$50\, e^{(0.04-0.03)\times 0.5} = \$50.25 \qquad \text{and} \qquad 50\, e^{0.04\times 0.5} = \$51.01$$

Any forward price between \$50.25 and \$51.01 is free of arbitrage: a band 76 cents wide where previously there was a point. A forward quoted at \$50.40 is not mispriced to anyone who must borrow the stock to act on it.

The band is reached only if no holder will sell: a shareholder who owns the stock outright still closes the gap at \$50.40 by selling at \$50, investing to receive \$51.01, and buying the forward. Heavily shorted stocks tend to be markets where few holders will, which is why the two conditions arrive together: the borrow is expensive *because* the free float is already lent, and what remains is held by people who will not sell. So on a stock where fees exceed 20% a year, a persistently "cheap" forward is offering a profit only to whoever already owns the stock and will part with it, and the price is telling you how few of those there are.

5.3 KNOWN INCOME

If the asset provides a known cash income (coupons on a bond, a declared dividend), the holder of the physical asset receives it and the holder of a forward does not. The forward price must be lower by the present value of that income.

Writing I for the present value of the income received during the contract's life,

$$F_0 = (S_0 - I)\, e^{rT} \tag{5.4}$$

The arbitrage is the same as before, with one adjustment: the trader who buys the asset and shorts the forward receives the income along the way and can invest it. Example 5.3 applies it to a coupon-bearing bond.

Example 5.3 A forward on a coupon-bearing bond

A bond trades at \$900. It pays a \$40 coupon in four months, and a nine-month forward contract is being priced. The risk-free rate is 4% for all maturities.

The present value of the coupon is

$$I = 40\, e^{-0.04\times 4/12} = 40 \times 0.986755 = \$39.47$$

so from (5.4)

$$F_0 = (900 - 39.47)\, e^{0.04\times 0.75} = 860.53 \times 1.030455 = \$886.74$$

The forward price is below the spot price, which looks odd until one recalls that the forward buyer forgoes a \$40 coupon to save nine months of financing on \$900—and here the coupon is worth more.

5.4 KNOWN YIELD

Sometimes income is better expressed as a proportion of the asset's price than as a fixed amount: a stock index paying a dividend yield, or a foreign currency earning interest. Writing q for the

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continuously compounded yield,

$$F_0 = S_0\, e^{(r-q)T} \tag{5.5}$$

The intuition is direct: holding the asset earns q and costs r to finance, so the forward price grows at the net rate $r - q$. If the yield exceeds the financing rate, the forward price is *below* spot.

Example 5.4 A stock index futures price

An equity index stands at 4,200. The dividend yield on the index is 1.8% and the three-month risk-free rate is 4.3%. From (5.5),

$$F_0 = 4200\, e^{(0.043-0.018)\times 0.25} = 4200 \times 1.006270 = 4{,}226.3$$

Index arbitrage. When the futures price departs from (5.5), the trade is to buy the index basket and sell futures, or the reverse. It is executed by computer, since it requires simultaneous transactions in hundreds of stocks, and the profit per unit is small. It is the staple of index market makers and of bank *delta-one* desks (the desks that run exposures moving one-for-one with the underlying, as against the option books of Part II), for whom a few basis points is a business only because the volume is enormous. Example 5.5 shows how little of a quoted gap survives the costs.

This is an institutional trade, and only an institutional one. Simultaneity across hundreds of stocks requires automated basket execution and direct market access; the short direction requires standing stock-borrow relationships; and the financing has to be arranged at institutional rates. Equation (5.5) is enforced by the algorithmic desks of dealers, index funds, and quantitative managers. Read it as a description of how the price comes to be disciplined, not as an opportunity available on a retail platform.

The position is then held to settlement rather than traded out of. Index futures settle in cash against a published index level, so the basket must be liquidated *at that level*, which is why expiration days carry a concentration of orders to trade at the closing price and why volume in the constituents spikes at the moment the settlement level is set. A basket sold a few minutes early has stopped being a hedge and become a directional position in five hundred stocks.

The two directions are not equally available. Buying the basket against short futures needs cash and a broker. Selling the basket against long futures needs a borrow in every constituent: the constraint of Section 5.1 reappearing in an equity market, as borrowing arrangements for hundreds of stocks, at fees that differ from stock to stock, on loans that can be recalled without notice. The no-arbitrage band is therefore wider below theoretical value than above it: a futures price that is too cheap can persist, where one equally rich is arbitraged away more readily. When index futures nonetheless sit slightly rich (above theoretical value), the cause is not this asymmetry but the demand for exposure in futures form and the cost of the balance sheet needed to meet it, which Section 5.6 takes up.

Example 5.5 Index arbitrage with costs

The futures in Example 5.4 are quoted at 4,240, against a theoretical 4,226.3, a gap of 13.7 index points. The trade is to buy the basket and short the futures.

Round-trip transaction costs (commissions, bid–ask spreads, and market impact across the basket) come to perhaps 8 index points. The net is 5.7 points, and on a contract sized at \$50 per index point,

$$5.7 \times \$50 = \$285 \text{ per contract}$$

Two lessons follow. The arbitrage does not enforce equality but a *band*: no-arbitrage requires

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only that the futures price stay within transaction costs of the theoretical value. And the width of that band depends on execution capability, so the same mispricing is an opportunity for one participant and not for another.

When index arbitrage becomes impossible, as in a market where the underlying stocks cannot be traded or where trading halts break the simultaneity the trade requires, the band widens without limit, and futures and cash can decouple entirely. This happened in October 1987, and it is the reason the relationship is a mechanism rather than a law.

5.5 VALUING AN EXISTING FORWARD CONTRACT

A forward contract is worth zero when entered. Later it is not, because the forward price has moved while the contract's delivery price K has not.

For a long forward position with delivery price K and current forward price F_0 for the same maturity, the value today, f, is

$$\boxed{f = (F_0 - K)\, e^{-rT}} \tag{5.6}$$

Derivation. Compare two portfolios. Portfolio A is the existing long forward at delivery price K; portfolio B is a newly entered long forward at the current forward price F_0, which by definition is worth zero.

At T both portfolios deliver the same asset, worth S_T. Portfolio A pays K for it and portfolio B pays F_0, so A's terminal value exceeds B's by exactly $F_0 - K$, whatever S_T turns out to be. Since this difference is certain, it can be discounted at the risk-free rate:

$$f_A - f_B = (F_0 - K)e^{-rT}$$

and since $f_B = 0$, equation (5.6) follows.

The step to notice is that the asset price cancels. The value of an existing forward does not depend on any model of the underlying, only on how far today's forward price has moved from the contract's own delivery price. This is why forward and swap books can be revalued from a curve alone. Example 5.6 values one such contract.

Example 5.6 Valuing a forward after the market moves

An investor is long a forward on a non-dividend-paying stock with a delivery price of \$48 and three months remaining. The stock now trades at \$52 and the risk-free rate is 4%.

The current three-month forward price is

$$F_0 = 52\, e^{0.04\times 0.25} = \$52.52$$

so from (5.6)

$$f = (52.52 - 48)\, e^{-0.04\times 0.25} = 4.52 \times 0.990050 = \$4.48$$

Equation (5.6) is the same structure as the swap valuation of Chapter 7: an off-market contract is worth the difference between its own rate and the current market rate, discounted. Chapter 7's $L(S - K)A$ is the annuitized version of the same idea.

5.6 STOCK INDEX FUTURES IN PRACTICE

Equation (5.5) treats the dividend yield as a known constant. For an index over a few months this is a reasonable approximation, since the constituent companies' dividends are largely announced in advance, but three practical points are worth making.

Dividends are lumpy, not continuous. The yield q is an approximation to a sequence of discrete payments concentrated in particular weeks. For contracts spanning a heavy dividend

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period the approximation matters, and desks price index futures off the actual projected dividend stream rather than a smoothed yield, returning in effect to equation (5.4).

Dividends are forecasts. Unlike a bond coupon, a dividend can be cut. In 2020 many companies suspended dividends within weeks, and index futures prices moved for that reason alone. The dividend assumption embedded in a futures price is a traded quantity in its own right: dividend futures and swaps exist precisely so it can be hedged.

Index futures are the cheapest way to hold the index. Because the contract requires only margin, a fund can obtain index exposure without buying the constituents. This is the same observation Chapter 6 makes about Treasury futures and duration, and it has the same consequence: steady demand for exposure in futures form tends to leave futures slightly rich to fair value. That richness is not an unclaimed riskless profit; it is what an arbitrageur is paid for committing the cash and the balance sheet to close the gap, which is why it survives (Section 5.15). The demand, not the borrow asymmetry of Section 5.4, decides which side the gap appears on: the asymmetry on its own would let a cheap futures price last longer than a rich one.

5.7 CURRENCY FORWARDS AND COVERED INTEREST PARITY

A foreign currency is an investment asset paying a known yield, the foreign risk-free rate r_f. Applying (5.5) with $q = r_f$ and S_0 quoted as units of domestic currency per unit of foreign currency,

$$F_0 = S_0\, e^{(r - r_f)T} \tag{5.7}$$

This is **covered interest parity**, and it says something economically strong: the forward exchange rate is determined entirely by the two interest rates, with no reference to anyone's view about where the currency is heading. A currency with a high interest rate trades at a forward *discount*, by exactly enough to offset the interest advantage.

How they are quoted. Equation (5.7) is stated in terms of "domestic" and "foreign," and the market does not use those words. A spot rate is quoted as units of the **quote** currency per one unit of the **base** currency: EUR/USD at 1.0850 is 1.0850 dollars per euro, and USD/JPY at 150.25 is 150.25 yen per dollar. Which currency is the base is pure convention: the euro, sterling, and the Australian and New Zealand dollars are quoted as base against the dollar, while the dollar is the base against most other currencies. Mapping this onto (5.7), *domestic is the quote currency and foreign is the base*, so for USD/JPY the domestic rate is the yen rate and the foreign rate is the dollar rate.

Forwards are not quoted outright. The interbank market quotes **forward points**: the difference between the forward and the spot rate, expressed in the last decimal place of the spot quote, which the client adds to spot to get the outright rate. Points are quoted rather than outright rates because they move far more slowly. Spot ticks continuously, while the interest differential driving the points does not, so a dealer can hold a points quote long after a spot quote has gone stale, and a foreign exchange (FX) swap, which is a spot leg against an offsetting forward leg, trades the points alone with no exposure to spot at all. The sign of the points carries the differential directly: negative points mean the base currency earns the higher interest rate.

Futures are quoted differently again, and for some pairs the other way up. Exchange-traded currency futures are quoted *outright*, in US dollars per unit of the other currency. For the euro that agrees with the interbank convention; for the yen it inverts it, so CME yen futures trade near 0.006650 dollars per yen where the interbank market quotes 150 yen per dollar. The two numbers are reciprocals of each other, and taking one for the other is a standard and expensive error.

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Example 5.7 A currency forward

The spot rate is 150 yen per dollar. The one-year dollar rate is 4.3% and the one-year yen rate is 0.5%. The quote currency is the yen and the base currency is the dollar, so the yen is domestic and the dollar foreign, and

$$F_0 = 150\, e^{(0.005-0.043)\times 1} = 150 \times 0.962713 = 144.41 \text{ yen per dollar}$$

A dealer would not quote 144.41. It would quote forward points of −5.59 yen against a spot of 150.00, or −559 points in a market where a **pip** (one unit in the last decimal place of the quote) is 0.01 yen, leaving the client to add them. The negative sign says at once that the base currency, the dollar, carries the higher interest rate.

The dollar is worth fewer yen forward than spot: a forward discount of 3.7%, or 3.8% continuously compounded, which exactly offsets its 3.8% interest advantage. An investor who borrows yen, converts to dollars, invests at 4.3%, and sells the dollars forward ends up exactly where borrowing yen at 0.5% would have left them. Borrowing the low-rate currency to invest in the high-rate one is the **carry trade**, and there is no riskless profit in it once the currency risk is hedged, and equation (5.7) is the statement of that.

An unhedged carry trade is a different matter entirely: it earns the interest differential and bears the currency risk, and whether that is attractive is an empirical question, not an arbitrage one.

Who uses them. Companies are the most familiar users. An American exporter invoicing in euros sells the euros forward to fix the dollar value of its receivables, and an importer buys its foreign currency forward to fix the cost of its payables; either way the forward removes the exchange rate risk between invoice and payment. The largest volume comes from asset managers, pension funds and insurers that hold foreign bonds and equities and hedge the currency back into their own, usually by rolling one- to three-month forwards or FX swaps, which removes the currency risk of the holding and leaves its local return. Banks use FX swaps to fund themselves: a bank whose deposits are in euros and whose loans are in dollars swaps the one into the other for a fixed term rather than borrowing dollars unsecured. When the demand to obtain dollars through FX swaps outruns the balance sheet dealers will commit to supplying them, the forward stops obeying equation (5.7), and that gap is the cross-currency basis that Section 7.8 prices.

And yet it fails. Chapter 7 shows that covered interest parity has been persistently violated since 2008, by tens of basis points, in the most liquid currency pairs in existence. That is a striking thing to say about an arbitrage relationship derived in three lines. Section 5.15 explains why, and the explanation is the most important qualification in this chapter.

Which foreign rate? Equation (5.7) contains r_f, and the choice is less obvious than it looks. It should be the rate at which the arbitrageur can actually invest the foreign currency for the contract's life, which for a bank executing the trade is not a government bond yield but a money market or repo rate, and may differ from the rate available to a different institution.

This is the same observation as Section 5.10: the relationship is enforced by a specific trade done by a specific participant, and the rate that belongs in the formula is the one that participant faces. Much of the apparent breakdown in covered interest parity dissolves, and the remainder becomes interpretable, once one stops asking what "the" risk-free rate is and starts asking what the marginal arbitrageur can actually borrow and lend at.

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5.8 COMMODITIES, STORAGE, AND CONVENIENCE YIELD

Commodities cost money to store, which is negative income. Writing u for the storage cost expressed as a continuous proportional rate,

$$F_0 = S_0\, e^{(r+u)T} \tag{5.8}$$

for an investment commodity such as gold.

Storage is often charged as an amount rather than a proportion of value: so many dollars an ounce a year for a vault, so many cents a bushel a month at a grain elevator. It is then handled exactly as known income was in Section 5.3, with the sign reversed. Writing U for the present value of the storage costs incurred during the contract's life,

$$F_0 = (S_0 + U)\, e^{rT} \tag{5.9}$$

The arbitrage is the basic one of Section 5.2: a trader who buys the commodity and sells it forward must also pay to store it, so the forward price has to repay both the purchase and the storage, with interest. Example 5.8 prices the same contract both ways.

Example 5.8 A gold forward

Gold trades at \$2,000 per ounce, the one-year rate is 4.3%, and storage and insurance cost 0.5% of value per year. From (5.8),

$$F_0 = 2000\, e^{(0.043+0.005)\times 1} = 2000 \times 1.049171 = \$2{,}098.34$$

Suppose instead the vault charges a flat \$10 an ounce, payable at the end of the year: the same 0.5% of today's price, stated as an amount. Its present value is

$$U = 10\, e^{-0.043\times 1} = 10 \times 0.957911 = \$9.58$$

so from (5.9)

$$F_0 = (2000 + 9.58)\, e^{0.043\times 1} = 2009.58 \times 1.043938 = \$2{,}097.88$$

Since the fee is paid at delivery, Ue^{rT} is the \$10 itself, and the forward price is simply the financed spot price, \$2,087.88, plus the fee. It sits slightly below the proportional figure because a proportional rate charges storage on a value that grows over the year, while a flat fee does not grow.

Consumption commodities and the convenience yield

For a consumption commodity the argument works in only one direction, and this is where the chapter's tidy equalities break down.

If $F_0 > S_0 e^{(r+u)T}$, the usual arbitrage applies: buy the commodity, store it, sell forward. Anyone can do this, so this inequality cannot persist.

If $F_0 < S_0 e^{(r+u)T}$, the reverse trade requires *selling the commodity short* (borrowing oil from someone who holds it). But holders of consumption commodities hold them because they intend to use them. A refinery will not lend its crude, at any price, if lending it means shutting down. The arbitrage is therefore unavailable, and we can conclude only that

$$F_0 \le S_0\, e^{(r+u)T} \tag{5.10}$$

or, with storage stated as a present value, $F_0 \le (S_0 + U)e^{rT}$.

The gap that (5.10) leaves open is filled by defining the **convenience yield** y, the benefit of holding the physical commodity rather than a claim on it, so that equality is restored:

$$F_0 = S_0\, e^{(r+u-y)T} \tag{5.11}$$

With storage as a present value, the same definition reads $F_0\, e^{yT} = (S_0 + U)e^{rT}$.

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It should be said plainly that y is not observed and not modeled: it is *defined* as whatever makes the equation hold. That does not make it useless, because what it measures is real: the value of having the physical commodity available when you need it, which rises when supply is tight.

Example 5.9 Backing out a convenience yield

Crude oil trades at \$80 and the one-year futures price is \$76. The risk-free rate is 4.3% and storage costs 3% a year. From (5.11),

$$76 = 80\, e^{(0.043+0.03-y)} \implies \ln(0.95) = 0.073 - y$$

so

$$y = 0.073 + 0.051293 = 12.43\%$$

A convenience yield of over 12% says the market places a large premium on having oil now rather than in a year. That is a statement about physical tightness (low inventories, constrained logistics), and traders read the term structure of convenience yields as a supply indicator rather than as a pricing artifact.

5.9 THE COST OF CARRY

All the results above are one formula. Define the **cost of carry** c as the cost of holding the asset: financing cost, plus storage, minus income. Then for an investment asset

$$F_0 = S_0\, e^{cT} \tag{5.12}$$

and for a consumption asset

$$F_0 = S_0\, e^{(c-y)T} \tag{5.13}$$

For a non-dividend stock $c = r$; for a stock index $c = r - q$; for a currency $c = r - r_f$; for a storable commodity $c = r + u$.

The cost of carry is a term structure, not a number. Writing c with no argument hides the fact that every ingredient varies with maturity: r is the zero rate to T, storage is contracted at different rates for different horizons, and the convenience yield moves most of all. What (5.12) needs at maturity T is $c(T)$, the *average* carry between now and T. It stands to the carry over each intervening month exactly as Chapter 4's zero rate stands to the forward rates underneath it (Figure 4.3), and inherits the same amplification: a mildly rising $c(T)$ implies a considerably steeper profile of carry between adjacent delivery months. This is why a real forward curve is not the single exponential of Figure 5.1, and why it can slope one way at the front and the other further out, which is what Figure 5.2 shows for four markets.

Figure 5.1 shows the two shapes this produces. When carry exceeds convenience yield the curve slopes upward (**contango**) and holding the physical asset is expensive relative to holding the futures. When convenience yield dominates the curve slopes downward (**backwardation**) and the market is paying a premium for immediate availability. Backwardation is the normal state for commodities in tight supply and, for a financial asset, where there is no convenience in holding the physical, arises only when the asset's income yield exceeds the financing rate, as for the dollar against the yen in Example 5.7.

What the curve looks like, market by market

Figure 5.1 draws the two shapes as though a market chooses between them. Real markets do not: each commodity has a characteristic curve, set by how cheaply it can be stored and how badly it is needed now. Table 5.1 summarizes, and Figure 5.2 draws four representative shapes.

Read that figure twice. The first reading is the shape. The second is the gap between the markers (the contracts that actually trade, at listed delivery months) and the dashed line, which

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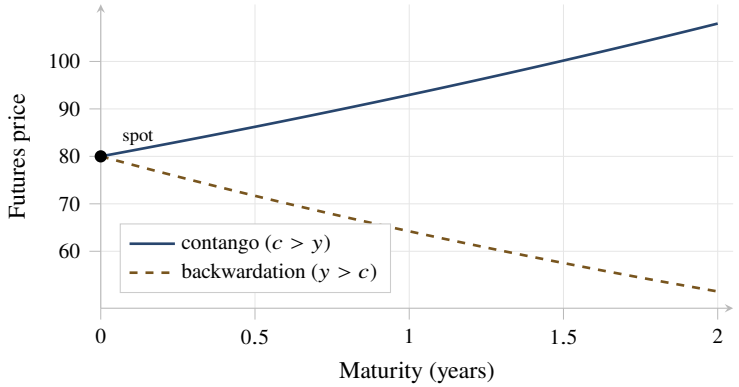


Figure 5.1: Forward curves when the cost of carry exceeds the convenience yield and when it does not. Constant carry, with the curvature exaggerated.

is (5.13) with a single convenience yield applied to every maturity. *A constant y fits none of them*: gold's markers sit a shade below the curve and drift further as its small convenience yield accumulates, crude's pull away as the front-month scarcity decays, gas swings with the season, and corn runs below full carry because commercial storage caps the slope. The convenience yield is a term structure, not a number, and the only honest way to read y off a market is one maturity at a time, which is what Example 5.10 does at a single point, and what Section 5.15 says cannot be pushed too far.

Table 5.1: Characteristic forward curve shapes.

Market	Typical shape	What sets it
Gold, precious metals	Contango, almost always	Above-ground stock is decades of mine supply, so $y \approx 0$ and the curve is the financing curve
Base metals	Mild contango; sharp backwardation in squeezes	Storable at moderate cost; shape tracks exchange inventory
Crude oil	Alternates; backwardated more often than not	y swings with inventory, and storage capacity is finite
Natural gas	Seasonal; neither label applies	Winter above summer; storage is costly and capacity-constrained
Grains, oilseeds	Rises within a crop year, breaks at the new crop	Harvest cycle; commercial storage cost caps the slope
Livestock	No carry relation	Not storable; animals must be fed and change in value
Electricity	No carry relation	Not storable at all; the forward is expectation plus risk premium

Gold: the pure carry case. Gold is the cleanest illustration of equation (5.12) in the whole commodity complex, because it barely behaves like a commodity at all. Roughly sixty years of mine production sits above ground in vaults and jewelry, so no plausible shortage can develop; the convenience yield of holding physical gold instead of a claim on it is close to zero. Storage and insurance run a fraction of a percent a year. What remains is financing cost, and the gold forward curve is therefore essentially the interest rate curve drawn in dollars per ounce.

This is why gold is almost never backwardated. When it briefly is, the cause is not a shortage of gold but a dislocation in the market for *immediately deliverable* gold in a particular location (refineries closed, flights grounded, bars of the wrong specification), and it resolves in

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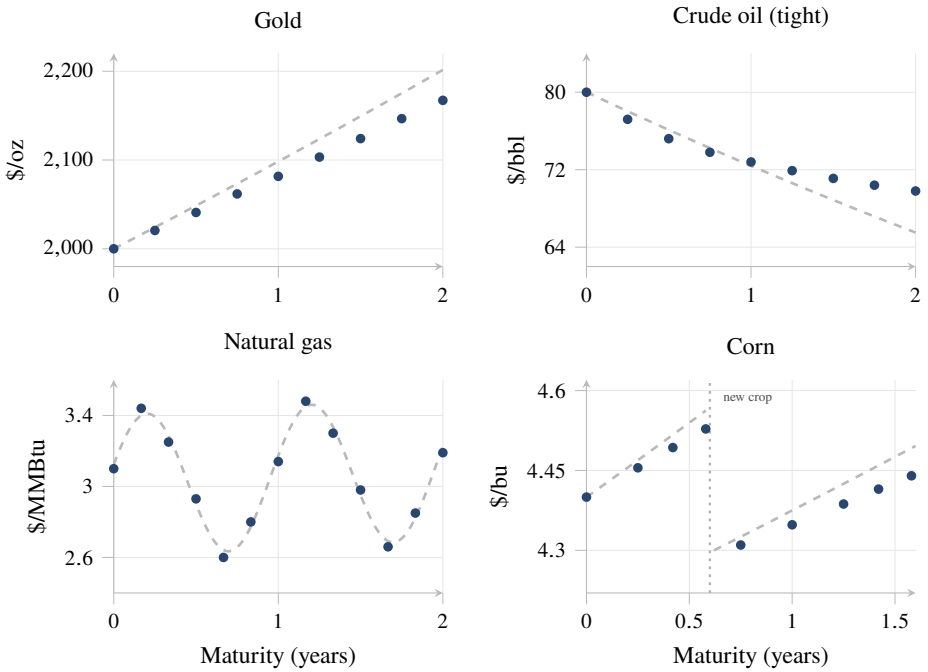


Figure 5.2: Four representative shapes. Markers are listed delivery months; the dashed line is what a constant convenience yield would give. Schematic.

weeks.

Example 5.10 Backing out the gold lease rate

Example 5.8 computed the full-carry forward price of gold, assuming no convenience yield: at a spot of \$2,000, financing of 4.3%, and storage and insurance of 0.5%, it came to \$2,098.34.

Suppose the one-year forward actually trades at \$2,081.60. The gap is small but it is not noise. The market reads it as a lease rate, defined as the financing rate less the forward premium on metal:

$$l = 0.043 - \ln\left(\frac{2081.60}{2000}\right) = 0.043000 - 0.039990 = 0.003010 = 0.30\%$$

Those 30 basis points are the **gold lease rate**: what a holder earns by lending metal to a bullion bank, and equivalently what a borrower pays to obtain metal now rather than in a year. It is a quoted market rate, and this residual calculation is how it is quoted. Storage does not enter it: the convention compares the forward premium with the financing rate alone. Solving (5.11) for the convenience yield adds storage back, $y = l + u = 0.30\% + 0.50\% = 0.80\%$.

The magnitude carries the lesson. A lease rate of 30 basis points, and a convenience yield under one percent, is what "essentially zero convenience yield" looks like in a real market, and it should be set beside the 12.43% of Example 5.9. The same equation governs both, and one parameter separates them by a factor of fifteen.

Crude oil: the regime-switching case. Oil spends most of its time backwardated. The reason is structural: refiners cannot run without crude, inventories are small relative to consumption, and a genuine shortfall is expensive in a way no financial hedge repairs. That sustains a positive convenience yield most of the time. But when demand falls or production overshoots, inventories build, the convenience yield collapses, and the curve flips into contango, and then the arithmetic of storage takes over.

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Suppose the front contract is \$42 and the twelve-month is \$54, a \$12 contango. A trader can buy physical crude, charter a tanker, and sell the deferred contract. A very large crude carrier holds about two million barrels; at \$60,000 a day the charter costs \$21.9 million a year, or about \$10.95 a barrel, and financing \$42 for a year at 2% adds \$0.84. Total carry is close to \$12, and the trade is marginal—which is the point. *Contango is capped near full carry*, because as soon as it exceeds the cost of storage, storage gets bought until it does not. This is not a hypothetical: floating storage on chartered tankers was a large and visible trade in 2009 and again in 2020.

The cap has a failure mode, and it is the one the next Case Study describes. The equilibrating mechanism works by *buying* storage. When there is no more storage to buy, nothing caps the contango, and prices do things that pricing formulas assuming a finite u cannot represent.

Natural gas: seasonal, so neither word applies. Gas demand is concentrated in winter heating and summer cooling, while production is roughly flat. Storage bridges the gap: gas is injected into underground facilities from April to October and withdrawn from November to March. The forward curve is therefore not monotone at all: it oscillates, with winter contracts above the summer contracts on either side, and calling it "in contango" or "in backwardation" is a category error.

The interesting point on such a curve is the March/April spread, which straddles the end of the withdrawal season. March is the last winter contract and April the first injection contract, so the spread is a direct bet on whether the winter ends with storage tight or comfortable. It is volatile enough to have earned the name *widow-maker*, and Section 5.14 returns to it: the largest hedge fund collapse of its era was a bet on this one spread.

Agriculture: carry within a crop year, a break across it. Grain arrives all at once at harvest and is consumed all year, so it must be stored, and within a crop year the curve normally rises at close to the cost of storing it. Grain markets have a name for the ceiling, **full carry** (interest plus commercial storage), and quote the spread as a percentage of it, as Example 5.11 does for corn.

What makes agricultural curves distinctive is that the carry relation *resets* at the new crop. A December corn contract and the following December contract are claims on different harvests, and no amount of storage links them, because grain stored from one harvest competes with grain that has not yet been planted. This is the discontinuity in the fourth panel of Figure 5.2, and it is why old-crop and new-crop contracts are quoted and traded as different markets.

Example 5.11 Reading a grain spread as a percentage of full carry

December corn trades at \$4.40 a bushel and March corn at \$4.55, both in the same crop year. Financing costs 5.0% and commercial elevator storage is 5 cents a bushel a month.

Over the three months from December to March, full carry is

$$\underbrace{4.40 \times 0.050 \times 0.25}_{\text{interest} = \$0.055} + \underbrace{3 \times 0.05}_{\text{storage} = \$0.150} = \$0.205$$

The market spread is $4.55 - 4.40 = \$0.15$, or

$$\frac{0.150}{0.205} = 73\% \text{ of full carry}$$

Read this as a signal about stocks. Near 100% of full carry, the market is paying anyone who stores grain their full cost, which it only needs to do when supplies are ample and someone must be induced to hold them. At 73% the market is paying most but not all of the cost: the 15-cent spread must first cover 5.5 cents of interest, so an elevator whose own storage cost for the three months is below the remaining 9.5 cents still profits by storing, one at full commercial rates does

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not, and grain moves out of the highest-cost storage first.

An *inverse* (March below December) reverses the message entirely. The market is then paying a premium to pull grain out of storage immediately, which is the agricultural version of backwardation and signals old-crop tightness. Storage cannot arbitrage it away, because the one thing carry cannot do is move grain backward in time.

Livestock and electricity: where the relation breaks. Two categories in Table 5.1 have no cost-of-carry relation at all, and they are worth naming because they mark the boundary of everything in this chapter.

Live cattle cannot be stored. An animal held for three months is not the same animal: it has been fed, at a cost, and has gained weight, changing what is delivered. Electricity is more extreme: outside limited battery and pumped-hydro capacity it cannot be stored for even an hour, so there is no trade that links today's price to next July's. In both markets the forward price is not spot compounded at anything. It is the market's expectation of the future spot price, plus a risk premium: the relationship of Section 5.13, standing alone because the arbitrage argument that usually dominates it is unavailable. This is why power forward curves show shapes no storable commodity ever displays, including negative prices in hours when the grid must dispose of generation.

Case Study 5.1 When storage runs out

On April 20, 2020, the expiring May contract for West Texas Intermediate (WTI) crude settled at −$37.63 a barrel. Buyers were paid to take oil.

Equation (5.8) assumes storage is available at a cost u. In April 2020, with demand collapsed by the pandemic and production still running, the tanks at the delivery point at Cushing, Oklahoma were effectively full. Longs approaching expiration faced physical delivery of oil they had nowhere to put, and could not roll into the next month fast enough. Storage was not expensive; it did not exist. The carry relation itself cannot produce a negative futures price, since it makes F a positive multiple of the spot price. What went negative was the spot value of a barrel at Cushing: a barrel that had to be taken with nowhere to put it had become a disposal cost, and the expiring contract, converging to that spot value, settled below zero. Section 36.1 returns to the episode.

Two lessons follow. First, the parameters in these formulas are not always finite, and a model that treats u as a small constant will be wrong in precisely the situations that matter. Second, this is what a *consumption* asset looks like when the consumption stops: the convenience yield collapsed to something deeply negative, and holding the physical was a liability, not a benefit.

The episode also broke a great deal of software. Option pricing models assuming lognormal prices cannot represent a negative underlying, exactly the problem Section 4.2 describes for interest rates (and Chapter 29 prices), and exchanges had to switch commodity options to normal models within days.

It broke retail products too, and more painfully. The largest losses fell on holders of a Bank of China product called Crude Oil Treasure, which gave retail clients exposure to WTI futures. Its client trading window closed hours before the exchange struck the settlement price, so holders could neither see the collapse nor exit it. And because the product was unleveraged (clients funded it in full), it had no margin close-out mechanism at all, on the reasoning that an unleveraged position cannot lose more than its principal. That reasoning is sound only while prices stay positive. Roughly sixty thousand investors lost their principal and finished owing the bank money, and Chinese

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regulators subsequently barred banks from offering such products to individuals.

The general lesson is worth more than the episode. Every product and every model contains variables that someone has quietly assumed to be nonnegative. It is worth asking which ones, and what happens if they are not.

5.10 THE IMPLIED REPO RATE

Practitioners rarely compute a theoretical forward price and compare it with the market. They invert the question: given the two prices, what return would the cash-and-carry trade actually earn? That return is the **implied repo rate**.

Buy the asset today at S_0, finance it, and deliver it at T for the invoice amount F_0, the sum the futures buyer pays on delivery. Ignoring intermediate income, the annualized return is

$$r_{\text{implied}} = \left(\frac{F_0}{S_0} - 1 \right) \times \frac{\text{days per year}}{\text{days to delivery}} \tag{5.14}$$

Comparing this with the rate at which the position can genuinely be financed, the *actual* repo rate, answers the question directly. If the implied repo rate exceeds the actual repo rate, the cash-and-carry trade earns more than it costs and should be done. If it is lower, the trade loses, and the reverse is indicated for anyone able to borrow the asset. Example 5.12 makes the comparison for a deliverable bond.

Example 5.12 Implied against actual repo

A bond has a cash price of 98.50. It can be delivered into a futures contract in 90 days for an invoice amount of 99.05. Financing in the repo market costs 4.30%.

From (5.14), using a 365-day year,

$$r_{\text{implied}} = \left(\frac{99.05}{98.50} - 1 \right) \times \frac{365}{90} = 0.0055838 \times 4.0556 = 2.26\%$$

The trade returns 2.26% and costs 4.30% to finance: a loss of 2.04% annualized. Buying the bond and selling the futures is not profitable here: the futures are *cheap* relative to the cash bond, and the indicated trade is the reverse, subject to being able to borrow the bond.

The implied repo rate is the practitioner's version of everything in this chapter. It restates the cost-of-carry relationship as a comparison between two interest rates, which is how the trade is actually assessed. Chapter 6 uses the same quantity under a different name: the net basis of Section 6.4 is the gross basis less carry, and it is negative precisely when the implied repo rate exceeds the actual one.

Which financing rate? Equation (5.14) compares against the *repo* rate, not the risk-free rate, and the distinction is the practical content of Section 5.15. The relevant rate is what this particular asset can be financed at by this particular participant, which depends on the asset's quality as collateral, on whether it is trading "special" in the repo market, and on the borrower's own standing. Two firms looking at the same two prices can correctly reach opposite conclusions.

5.11 ROLLING A POSITION AND ROLL YIELD

Futures contracts expire; an investor wanting continuous exposure must roll, selling the expiring contract and buying the next. The shape of the curve in Figure 5.1 then has a direct and often underappreciated effect on returns, which Example 5.13 isolates by holding the spot price fixed.

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Example 5.13 The cost of rolling in contango

An investor wants continuous exposure to a commodity trading at \$80 spot. The one-month futures contract trades at \$80.50, a market in mild contango.

The investor buys the one-month contract at \$80.50. Suppose the spot price does not move at all over the month. By Chapter 2's convergence argument, the contract must approach \$80 as it expires. The investor therefore loses \$0.50, sells at \$80, and buys the new one-month contract at \$80.50 to continue.

Repeating twelve times, the position loses

$$12 \times \$0.50 = \$6.00$$

or 7.5% of the initial \$80—on a commodity whose price never changed.

In backwardation the arithmetic reverses. With the one-month contract at \$79.50, the same investor gains \$0.50 each month and earns 7.5% a year from a flat spot price.

This is the **roll yield**, and for long-horizon commodity exposure it frequently dominates the spot price move entirely. It is the reason exchange-traded products tracking commodity futures can fall over years in which the underlying commodity rose, and why the offering documents of such products describe their roll methodology at length.

The roll yield is not a market inefficiency and there is no arbitrage in it. It is the cost of carry, collected month by month by whoever holds the physical and paid month by month by whoever wants exposure without holding it. What flows is the *net* carry $c - y$, so the direction can reverse. In Example 5.9 the convenience yield exceeds financing and storage: the holder of physical oil takes that yield in kind, and the futures price, set below spot, hands the futures investor a positive roll yield instead.

5.12 FORWARD VERSUS FUTURES PRICES

We have used "forward price" and "futures price" interchangeably. Chapter 2 explained why they are not the same instrument: futures are settled daily, forwards at maturity.

The consequence is precise. If interest rates were certain, the two prices would be identical: daily settlement would move cash around but the amounts would be known. When rates are stochastic the timing matters, and the direction depends on the correlation between the asset price and interest rates.

Consider an asset *positively* correlated with rates. A trader long futures makes money when the asset rises, and the gain arrives as variation margin at a moment when rates are high, so it is reinvested well. Losses arrive when rates are low and are financed cheaply. Being long futures is therefore better than being long a forward, and the futures price must be higher to compensate.

For an asset negatively correlated with rates the argument reverses and the futures price is lower.

The magnitude is negligible for short-dated contracts on assets weakly correlated with rates, which is why the distinction is usually ignored. It is emphatically not negligible for interest rate contracts, where the asset *is* the rate and the correlation is one. Chapter 30 develops this into the convexity adjustment, which reaches tens of basis points, and Chapter 6 shows the consequences for curve construction. Section 28.7 states the same result in one line: daily settlement makes the money market account the futures contract's numeraire, so the futures price is a martingale under the risk-neutral measure while the forward price is one under the T-forward measure.

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5.13 FUTURES PRICES AND EXPECTED SPOT PRICES

We come finally to the question every newcomer asks: is the futures price the market's forecast of the future spot price?

The answer is no, and the reason is instructive.

Consider a speculator who takes a long futures position, holds it to delivery, and sets aside the present value of the futures price, $F_0 e^{-rT}$, in a risk-free investment. At T the investment pays F_0, which buys the asset under the contract, so the speculator ends with an asset worth S_T. The outlay today is certain and is discounted at r; the payoff is risky and is discounted at k, the rate appropriate to the asset's risk. Under standard asset pricing the present value of the whole position must be zero:

$$-F_0\, e^{-rT} + E[S_T]\, e^{-kT} = 0$$

so that

$$F_0 = E[S_T]\, e^{-(k-r)T} \tag{5.15}$$

If the asset has no **systematic risk** (risk that moves with the market as a whole and so cannot be diversified away), then $k = r$ and (5.15) gives $F_0 = E[S_T]$. If it has positive systematic risk (most assets), then $k > r$ and $F_0 < E[S_T]$: the futures price sits below the expected spot price, and the speculator earns the difference as compensation for bearing risk.

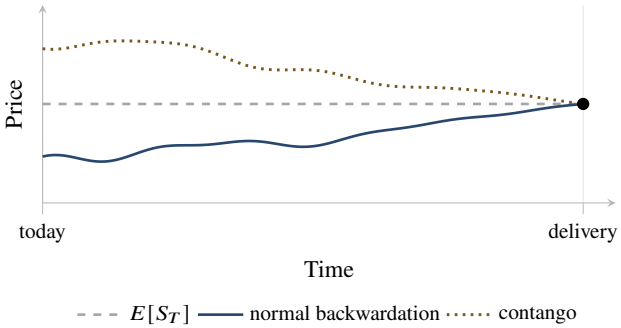


Figure 5.3: The futures price drifting toward the expected spot price as delivery approaches. Illustrative paths; the drift is the claim, not the route.

Figure 5.3 shows what this implies over time. If $F_0 < E[S_T]$, the futures price must drift *upward* on average as delivery approaches, since it must converge to S_T. Keynes called this **normal backwardation**, and explained it by hedging pressure: if producers are net short (as they typically are, hedging future output), speculators must be net long, and will require compensation. Where hedgers are net long instead, the argument reverses and $F_0 > E[S_T]$, which Keynes called **contango**.

Two warnings on terminology. The words are used in two different senses and the ambiguity causes real confusion. Section 5.9 used contango and backwardation to describe the *shape of the forward curve* today, which is observable. Keynes used them to describe the relationship between the futures price and the *expected* spot price, which is not observable, since no one sees $E[S_T]$. Most market usage means the first. Statements that a market "in contango" implies traders expect prices to rise confuse the two, and are wrong: the shape of the curve is determined by carry and convenience, not by expectations.

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5.14 TRADING THE SHAPE OF THE CURVE

Two results in this chapter describe money changing hands. Section 5.11 showed that the shape of the curve determines a roll yield, and Section 5.13 showed that a risk premium is compensation somebody is paying. Both are therefore things a speculator can try to earn, and three expressions are common.

Calendar spreads. Buying one delivery month and selling another is a direct bet on the shape of the curve, with no outright exposure to the level of the price. If the front contract is expected to richen relative to the second, because inventories are tightening, the trade is long the front and short the deferred. Section 2.3 showed why this is efficient: the two legs offset, so the margin requirement is a fraction of the outright, and Example 2.3's calendar spread required a twentieth of the gross amount.

Case Study 5.2 Amaranth and the widow-maker

In September 2006 Amaranth Advisors, a multi-strategy hedge fund managing about $9.2 billion, lost roughly $6.6 billion in a matter of weeks and closed. Almost all of it came from natural gas calendar spreads: principally long March against short April, the spread of Section 5.9 that straddles the end of the withdrawal season.

The position was not obviously foolish. A cold winter drains storage, March scarcity bids the last winter contract above the first injection contract, and the spread widens. It had done so before. But the 2005–06 winter had been mild, storage ended the season full, and the 2006 hurricane season passed without disrupting Gulf production. The spread collapsed instead of widening.

What turned a bad trade into a fatal one was size. Amaranth's positions were large enough that the Senate Permanent Subcommittee on Investigations later found it had at times held on the order of 40% of the open interest in NYMEX winter contracts. A position that large cannot be exited: every attempt to sell moved the spread further against the remaining position, because there was no one on the other side but the firm itself. The fund's positions were eventually transferred in bulk to JPMorgan and Citadel at a steep discount.

Three lessons follow, each pointing at a different chapter. **Low margin is not low risk.** Spread positions receive the margin offsets of Section 2.3 precisely because the legs hedge each other against parallel moves, which is exactly what lets a trader accumulate a notional position that a directional book could never finance. **Liquidity is part of the position.** A trade you cannot exit is a different trade from the one you put on. **Margin arrives before the thesis resolves.** The variation margin mechanism of Section 2.3 does not wait to find out whether you were right about the winter.

The episode also had a regulatory afterlife. Amaranth had shifted positions from NYMEX to the then-unregulated Intercontinental Exchange (ICE) to escape NYMEX's position accountability levels (the position sizes above which the exchange can demand an explanation and order a position cut), and closing that gap was a direct motivation for extending Commodity Futures Trading Commission (CFTC) oversight to economically equivalent contracts traded off-exchange.

Carry harvesting. A systematic strategy holds long positions in backwardated markets and short positions in contangoed ones, collecting the roll yield of Section 5.11 across many commodities. This is the commodity version of a carry trade, and like all carry trades it earns a steady return punctuated by sharp losses when the curve inverts.

Taking the other side of hedging pressure. If Keynes was right that producers' hedging demand pushes F_0 below $E[S_T]$, a speculator who simply holds long positions in commodities

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where producers dominate should earn the difference.

Does the premium exist? The empirical answer is that it is real but neither large nor reliable. Studies find returns consistent with a risk premium in some commodities and periods and not others, and the premium appears to move with the balance of hedging pressure, which is itself unstable, since whether producers or consumers dominate a market changes with industry structure.

This is a genuine risk premium and not an arbitrage. It is compensation for bearing a risk that someone else wanted to shed, it is earned on average and lost in specific episodes, and it disappears when hedging demand switches direction.

Why a hedger should care. This is not a digression from hedging; it is the same fact seen from the other side. The speculator's expected return is the hedger's expected cost. A producer hedging in a persistently backwardated market where producers dominate is systematically paying a premium to transfer risk, and that premium is the price of the insurance, not evidence that hedging is a mistake.

It also disposes of a common confusion. Contango is not "bad" and backwardation is not "good." Example 3.6 showed a short hedger gaining from rolling in contango, and Section 5.11 showed a long investor losing from it in the same market. Any statement about the shape of the curve that does not specify which side of the market one is on is empty.

Chapter 36 develops these strategies properly for energy and commodity markets, where the term structure carries the most information.

5.15 THE LIMITS OF ARBITRAGE

Every result in this chapter rests on the assumption stated in Section 5.1: that participants can borrow and lend freely at the risk-free rate and will take arbitrage opportunities when they see them. It is time to examine what that assumption costs.

The arbitrage in Example 5.1 requires borrowing \$50 and holding a stock for six months. That is a balance sheet position. Post-crisis capital rules (the leverage ratio of Section 1.12 in particular) make holding it expensive, and the expense does not shrink because the position is hedged.

This is not an abstraction, and the next two chapters meet it three times:

- **Covered interest parity** (Section 5.7) has been violated continuously since 2008. Chapter 7 shows the deviation is the price of the balance sheet required to close it.
- **Swap spreads** (Chapter 7) turned negative, offering an apparently riskless \$811,100 on a \$100 million ten-year position, which sat available for years because the trade requires financing a bond.
- **The Treasury cash–futures basis** of Chapter 6 is the arbitrage of Example 5.1 applied to government bonds. It persists, and is harvested by funds willing to run fifty times leverage—which is what the balance sheet constraint looks like when someone is prepared to bear it.

Case Study 5.3 Arbitrage requires a balance sheet

The three dislocations above are usually discussed separately (one in FX, one in rates, one in Treasuries) and are usually presented as puzzles. They are the same phenomenon.

In each case the textbook arbitrage exists and is riskless in the sense this chapter means. In each case executing it requires holding an asset and financing it. And in each

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case post-crisis regulation made that holding costly in a way that does not depend on the position's risk, so the cost cannot be hedged away.

The consequence is a genuine change in what "no-arbitrage" means. Before 2008 it was reasonable to treat these relationships as identities, because the intermediaries who enforced them faced negligible costs for doing so. They are now better understood as relationships that hold up to a spread, where the spread is the price of intermediation. That price is highest when dealers have the least balance-sheet capacity to spare, at reporting dates and in periods of stress, and those are the times when the dislocations are largest.

For a reader of this chapter, the practical implication is not to distrust the formulas. They remain the right starting point and the right way to think. It is to be precise about what enforces them, and to expect the enforcement to weaken under stress rather than to strengthen.

One consequence deserves naming here, because this chapter has used r throughout without saying whose rate it is. For a *collateralized* derivative the question has a definite answer, and it is not a matter of judgment: the discount rate is the rate paid on the collateral, fixed by the collateral agreement rather than chosen by the valuer. Section 7.4 derives it. Where a position is collateralized in its own currency, this chapter's r is that rate; where it is not (an uncollateralized trade, or collateral posted in another currency), the difference becomes an explicit valuation adjustment in Chapter 9.

SUMMARY

For an investment asset, arbitrage in both directions forces $F_0 = S_0 e^{rT}$, and no view about the asset is needed. Income lowers the forward price, whether it is known in amount, $F_0 = (S_0 - I)e^{rT}$, or as a yield, $F_0 = S_0 e^{(r-q)T}$; storage raises it; a currency is an asset yielding the foreign rate, so a hedged carry trade earns nothing. An existing forward is worth $(F_0 - K)e^{-rT}$, with no model of the asset required.

The relation is an equality only where holders are willing to sell. A borrowing fee opens a band when they are not, and for a consumption commodity they never are, so only an upper bound survives and the convenience yield closes the gap. Everything reduces to the cost of carry, itself a term structure: carry above the convenience yield gives contango, below it backwardation. April 2020 showed what happens when storage is unavailable rather than merely costly.

Practitioners run the relation in reverse: the implied repo rate says whether a cash-and-carry pays, and the cost of rolling in contango is the cost of carry, paid by whoever wants exposure without the physical. Neither the forward nor the futures price is a forecast; the futures price sits below the expected spot price when the asset carries systematic risk, so contango does not mean prices are expected to rise. Finally, arbitrage needs a balance sheet, which post-crisis regulation made expensive, so the formulas hold only up to a spread, and the spread is widest when dealers can least afford to close it.

FURTHER READING

Working (1949) is the origin of the theory of storage and states why a forward curve slopes the way it does; Keynes's *A Treatise on Money* (1930, vol. 2) is where normal backwardation comes from, and the distinction between the two ideas is worth getting straight. Cornell and Reinganum (1981) is the standard early test of forward against futures prices. On why the arbitrage this chapter relies on can fail, Shleifer and Vishny (1997) is the general argument

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and Du, Tepper, and Verdelhan (2018) the specific one, documenting deviations from covered interest parity that ought not to exist.

PRACTICE QUESTIONS

5.1 A non-dividend-paying stock trades at \$72 and the four-month risk-free rate is 3.6%. Compute the four-month forward price, and describe the arbitrage if the forward is quoted at \$74, giving the profit per share.

5.2 A stock will pay a \$1.50 dividend in two months and \$1.50 in five months. It trades at \$64 and the six-month rate is 4%, flat. Compute the six-month forward price.

5.3 An index stands at 3,800 with a dividend yield of 2.1% and a six-month rate of 4.0%. Compute the six-month futures price, state what it would be if the dividend yield exceeded the interest rate and why, and explain why that yield is a forecast rather than a known quantity.

5.4 An investor is short a forward on a non-dividend-paying stock with a delivery price of \$95 and five months remaining. The stock trades at \$88 and the rate is 3.5%. Value the position.

5.5 The spot rate is 1.08 dollars per euro, the one-year dollar rate 4.2% and the euro rate 2.6%. Compute the one-year forward rate, state which currency is at a forward discount and why, and explain why a hedged carry trade earns nothing while an unhedged one is a different proposition entirely.

5.6 Copper trades at \$4.20 per pound with a one-year futures price of \$4.05, a rate of 4.3%, and storage costs of 1.5%. Compute the implied convenience yield and interpret it. Repeat the carry calculation for silver at \$29 per ounce with storage of 0.4% and a rate of 4.3%, and explain why the two cases require different arguments.

5.7 Explain the difference between the two uses of the word “contango” in this chapter, and why it is wrong to say that a market in contango expects prices to rise.

5.8 A stock trades at \$120 with a three-month rate of 4% and a borrowing fee of 5% a year. Compute the no-arbitrage band for the three-month forward price, and explain why a quote inside the band is not a mispricing.

5.9 A bond has a cash price of 101.20 and can be delivered into a futures contract in 60 days for an invoice amount of 101.90. Compute the implied repo rate on a 365-day basis. If the bond can be financed at 4.10%, should the cash-and-carry trade be done?

5.10 Derive the value of a long forward contract by comparing two portfolios. Explain why the underlying asset price does not appear in the result, and what this permits operationally.

5.11 A commodity trades at \$60 spot with the one-month futures price at \$60.75, and an investor holds continuous long exposure by rolling monthly. If the spot price is unchanged after a year, what is the return? Repeat for a one-month futures price of \$59.40, and identify who receives what the futures investor pays.

5.12 Silver trades at \$29 per ounce, and storage and insurance cost \$0.12 per ounce per quarter, payable at the start of each quarter. The interest rate is 4.3% for all maturities. Compute the nine-month futures price. If the nine-month futures contract trades at \$30.40, describe the arbitrage and give the profit per ounce.

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FURTHER QUESTIONS

5.13 Explain why the arbitrage argument for a consumption commodity works in only one direction, and derive the resulting inequality. Is the convenience yield a model, a measurement, or a definition? Defend your answer.

5.14 In April 2020 the expiring WTI contract settled below zero. Explain which assumption in equation (5.8) failed, what it implies about the convenience yield in that month, and what pricing models had to be changed as a result.

5.15 Explain why forward and futures prices differ when interest rates are stochastic, deriving the direction of the difference from the correlation between the asset and rates. Why is the effect negligible for equity index futures and material for interest rate futures?

5.16 A trader observes that index futures are 20 points above their theoretical value. Set out the trade, identify every cost and risk it involves, and explain what would have to be true for the trade to be genuinely riskless. Under what market conditions would you decline to do it?

5.17 A stock is impossible to borrow at any price. Explain why this alone does not put the forward price below $S_0 e^{rT}$, state the condition on the shareholder register under which it does, and say why that condition and an expensive borrow tend to arrive together.

5.18 Explain why the no-arbitrage band around an index futures price is wider below theoretical value than above it. State what a participant needs to close the gap in each direction, and explain why, although index futures are more often rich than cheap, the asymmetry is not the reason.

5.19 A dealer quotes USD/JPY spot at 148.50 and one-year forward points of -480. Give the outright one-year forward rate, identify the base and quote currencies, say which currency carries the higher interest rate and how you know, and state roughly what a CME yen futures contract for the same date would be quoted at.

5.20 Equation (5.12) writes the cost of carry as a single number c. Identify which of its components vary with maturity, explain what $c(T)$ must therefore mean, and state the analogy with the zero and forward rates of Chapter 4. What does the analogy predict about the carry implied between two adjacent delivery months when $c(T)$ is rising gently?

5.21 Covered interest parity, negative swap spreads, and the Treasury cash–futures basis are usually discussed separately. Argue that they are one phenomenon. What does your answer imply about how the no-arbitrage results of this chapter should be taught and used?

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